

Data Science Internship|Level 1- Task 1

Objective :-

The objective of this task is to explore the dataset and perform preprocessing by identifying missing values, understanding the structure of the data, and analysing the distribution of the target variable "Aggregate rating".

Dataset Overview :-

The dataset contains information about restaurants, including details such as location, cuisines, ratings, and cost.

- Total Rows: 9551
- Total Columns: 21

Code (Important) :-

```
1  import pandas as pd
2
3  df = pd.read_csv("Dataset.csv")
4
5  # Clean column names
6  df.columns = df.columns.str.strip()
7
8  # ✅ FIX (correct way)
9  df['Cuisines'] = df['Cuisines'].fillna("Unknown")
10
11 # 🔍 Check again
12 print(df['Cuisines'].isnull().sum())
13
14 # Check full info again
15 df.info()
16
17 df['Aggregate rating'].describe()
18 df['Aggregate rating'].value_counts()
19 print(df['Aggregate rating'].describe())
20 print(df['Aggregate rating'].value_counts())
```

Data Exploration :-

The dataset was explored using functions like `.head()`, `.shape()`, and `.info()`. These functions helped in understanding the structure, data types, and completeness of the dataset.

Output :-

```
PS C:\Users\ASUS\Downloads\my code> python -u "c:\Users\ASUS\Downloads\my code\PYTHON PROGRAMMING\task1.py"
0
<class 'pandas.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   Restaurant ID          9551 non-null  int64  
1   Restaurant Name         9551 non-null  str     
2   Country Code           9551 non-null  int64  
3   City                   9551 non-null  str     
4   Address                9551 non-null  str     
5   Locality               9551 non-null  str     
6   Locality Verbose       9551 non-null  str     
7   Longitude              9551 non-null  float64 
8   Latitude               9551 non-null  float64 
9   Cuisines               9551 non-null  str     
10  Average Cost for two   9551 non-null  int64  
11  Currency               9551 non-null  str     
12  Has Table booking      9551 non-null  str     
13  Has Online delivery    9551 non-null  str     
14  Is delivering now      9551 non-null  str     
15  Switch to order menu   9551 non-null  str     
16  Price range            9551 non-null  int64  
17  Aggregate rating       9551 non-null  float64 
18  Rating color           9551 non-null  str     
19  Rating text            9551 non-null  str     
20  Votes                 9551 non-null  int64  
dtypes: float64(3), int64(5), str(13)
memory usage: 1.5 MB
count    9551.000000
mean      2.666370
std       1.516378
min       0.000000
25%       2.500000
50%       3.200000
75%       3.700000
max       4.900000
Name: Aggregate rating, dtype: float64
Aggregate rating
0.0    2148
3.2     522
3.1     519
3.4     498
3.3     483
3.5     480
3.0     468
3.6     458
3.7     427
3.8     400
2.9     381
3.9     335
2.8     315
4.1     274
4.0     266
2.7     250
4.2     221
2.6     191
4.3     174
4.4     144
2.5     110
4.5      95
2.4      87
4.6      78
1.9       2
1.8       1
Name: count, dtype: int64
PS C:\Users\ASUS\Downloads\my code>
```

Handling Missing Values :-

Missing values were checked using: `df.isnull().sum()`

It was found that the "Cuisines" column had 9 missing values.

These missing values were handled using: `df['Cuisines'] = df['Cuisines'].fillna("Unknown")`

Data Type Verification :-

All columns were checked for correct data types using: `df.dtypes`

The dataset contains:

- Numerical data (int64, float64)
- Categorical data (string/object)

No major data type conversions were required.

Target Variable Analysis (Aggregate Rating) :-

The target variable "Aggregate rating" was analysed using:

`df['Aggregate rating'].describe()`

`df['Aggregate rating'].value_counts()`

Conclusion :-

The dataset was successfully explored and pre-processed to ensure it is suitable for further analysis. It contains a total of 9551 rows and 21 columns, representing detailed information about restaurants.

During the preprocessing stage, missing values were identified in the "Cuisines" column, with a total of 9 missing entries. These were handled by replacing them with the value "Unknown", ensuring no data loss and maintaining dataset consistency.

The data types of all columns were examined and found to be appropriate, with numerical and categorical features correctly defined. No major data type conversions were required. Further analysis of the target variable "Aggregate rating" revealed that the average rating is approximately 2.67, with most values lying between 3.0 and 4.0. A significant portion of the dataset contains restaurants with a rating of 0.0, indicating potential unrated entries. Overall, the dataset has been cleaned, structured, and prepared effectively, making it ready for further descriptive and predictive analysis.

Data Science Internship | Level 1 – Task 2

Objective :-

The objective of this task is to perform descriptive analysis on the dataset by calculating statistical measures for numerical features and exploring the distribution of categorical variables such as city and cuisines.

Code :-

```
22 # -----TASK - 1 ENDS HERE-----
23
24 # -----TASK - 2 STARTS HERE-----
25
26 print("\n--- Numerical Summary ---")
27 num_df = df.select_dtypes(include=['int64', 'float64'])
28
29 print("Mean:\n", num_df.mean())
30 print("\nMedian:\n", num_df.median())
31 print("\nStandard Deviation:\n", num_df.std())
32
33 print("\n--- Top Cities ---")
34 print(df['City'].value_counts().head(10))
35
36 print("\n--- Top Cuisines ---")
37 df['Cuisines'] = df['Cuisines'].str.split(', ')
38 cuisines_exploded = df.explode('Cuisines')
39 print(cuisines_exploded['Cuisines'].value_counts().head(10))
40
```

Output :-

```
--- Numerical Summary ---
Mean:
  Restaurant ID      9.051128e+06
  Country Code      1.836562e+01
  Longitude         6.412657e+01
  Latitude          2.585438e+01
  Average Cost for two 1.199211e+03
  Price range       1.804837e+00
  Aggregate rating    2.666370e+00
  Votes             1.569097e+02
dtype: float64

Median:
  Restaurant ID      6.004089e+06
  Country Code      1.000000e+00
  Longitude         7.719196e+01
  Latitude          2.857047e+01
  Average Cost for two 4.000000e+02
  Price range       2.000000e+00
  Aggregate rating    3.200000e+00
  Votes             3.100000e+01
dtype: float64

Standard Deviation:
  Restaurant ID      8.791521e+06
  Country Code      5.675055e+01
  Longitude         4.146706e+01
  Latitude          1.100794e+01
  Average Cost for two 1.612118e+04
  Price range       9.056088e-01
  Aggregate rating    1.516378e+00
  Votes             4.301691e+02
dtype: float64
```

```
--- Top Cities ---
City
New Delhi      5473
Gurgaon        1118
Noida          1080
Faridabad      251
Ghaziabad      25
Ahmedabad      21
Amritsar        21
Bhubaneshwar    21
Guwahati        21
Lucknow         21
Name: count, dtype: int64

--- Top Cuisines ---
Cuisines
North Indian   3960
Chinese        2735
Fast Food      1986
Mughlai        995
Italian        764
Bakery         745
Continental    736
Cafe           703
Desserts       653
South Indian   636
Name: count, dtype: int64
```

Statistical Analysis of Numerical Columns :-

Basic statistical measures such as mean, median, and standard deviation were calculated for all numerical columns.

Key Observations:

- The average aggregate rating is approximately 2.67, while the median is 3.2, indicating that most restaurants have moderate ratings.
- The average cost for two is around 1199, but the median is 400, showing that the data is highly skewed due to some expensive restaurants.
- The average number of votes is 156, while the median is 31, indicating that a few restaurants receive significantly higher engagement.
- The price range has a mean of 1.8, suggesting that most restaurants fall in the lower to mid price categories.

Analysis of Categorical Variables :-

Top Cities

The dataset shows that the majority of restaurants are concentrated in a few cities:

- New Delhi has the highest number of restaurants (5473)
- Followed by Gurgaon (1118) and Noida (1080)
- Other cities like Faridabad and Ghaziabad have significantly fewer entries

👉 This indicates that the dataset is heavily concentrated in Delhi NCR region

Top Cuisines

After splitting multiple cuisines, the most popular cuisines are:

- North Indian (3960) – most dominant
- Chinese (2735)
- Fast Food (1986)
- Followed by Mughlai, Italian, Bakery, and Continental

👉 This shows that North Indian cuisine is the most popular, followed by a mix of international and fast food options.

Conclusion :-

The descriptive analysis reveals important insights about the dataset:

- Numerical features show skewness, especially in cost and votes, indicating the presence of outliers.
- The dataset is geographically concentrated, with most restaurants located in New Delhi and nearby cities.
- North Indian cuisine dominates, followed by Chinese and Fast Food, reflecting customer preferences.

Overall, the dataset provides meaningful insights into restaurant distribution, pricing, and customer preferences, which can be useful for further analysis and modelling.

Data Science Internship|Level 1 – Task 3

Objective :-

The objective of this task is to analyse the geographical distribution of restaurants using latitude and longitude data and to explore the relationship between location and restaurant ratings.

Code :-

```
42 # -----TASK - 3 STARTS HERE-----
43
44 import matplotlib.pyplot as plt
45 # ----- PART 1: Restaurant Locations -----
46 print("\n--- Restaurant Locations (Scatter Plot) ---")
47 plt.figure(figsize=(8,6))
48 plt.scatter(df['Longitude'], df['Latitude'])
49 plt.title("Restaurant Locations")
50 plt.xlabel("Longitude")
51 plt.ylabel("Latitude")
52 plt.show()
53 # ----- PART 2: Top Cities Distribution -----
54 print("\n--- Top 10 Cities with Most Restaurants ---")
55 top_cities = df['City'].value_counts().head(10)
56 print(top_cities)
57 plt.figure(figsize=(8,6))
58 top_cities.plot(kind='bar')
59 plt.title("Top 10 Cities with Most Restaurants")
60 plt.xlabel("City")
61 plt.ylabel("Number of Restaurants")
62 plt.xticks(rotation=45)
63 plt.show()
64 # ----- PART 3: Location vs Rating -----
65 print("\n--- Longitude vs Aggregate Rating ---")
66 plt.figure(figsize=(8,6))
67 plt.scatter(df['Longitude'], df['Aggregate rating'])
68 plt.title("Longitude vs Aggregate Rating")
69 plt.xlabel("Longitude")
70 plt.ylabel("Aggregate Rating")
71 plt.show()
72 print("\n--- Latitude vs Aggregate Rating ---")
73 plt.figure(figsize=(8,6))
74 plt.scatter(df['Latitude'], df['Aggregate rating'])
75 plt.title("Latitude vs Aggregate Rating")
76 plt.xlabel("Latitude")
77 plt.ylabel("Aggregate Rating")
78 plt.show()
79
```

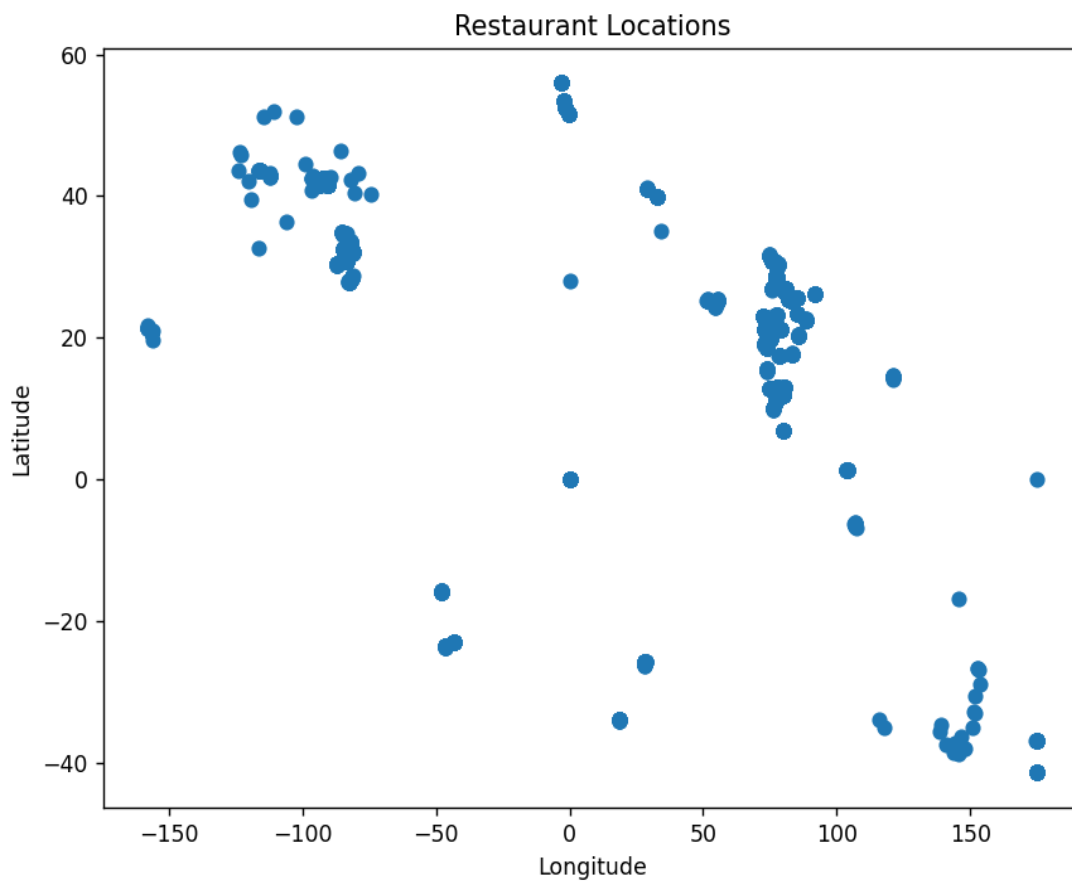
Output :-

```
--- Restaurant Locations (Scatter Plot) ---

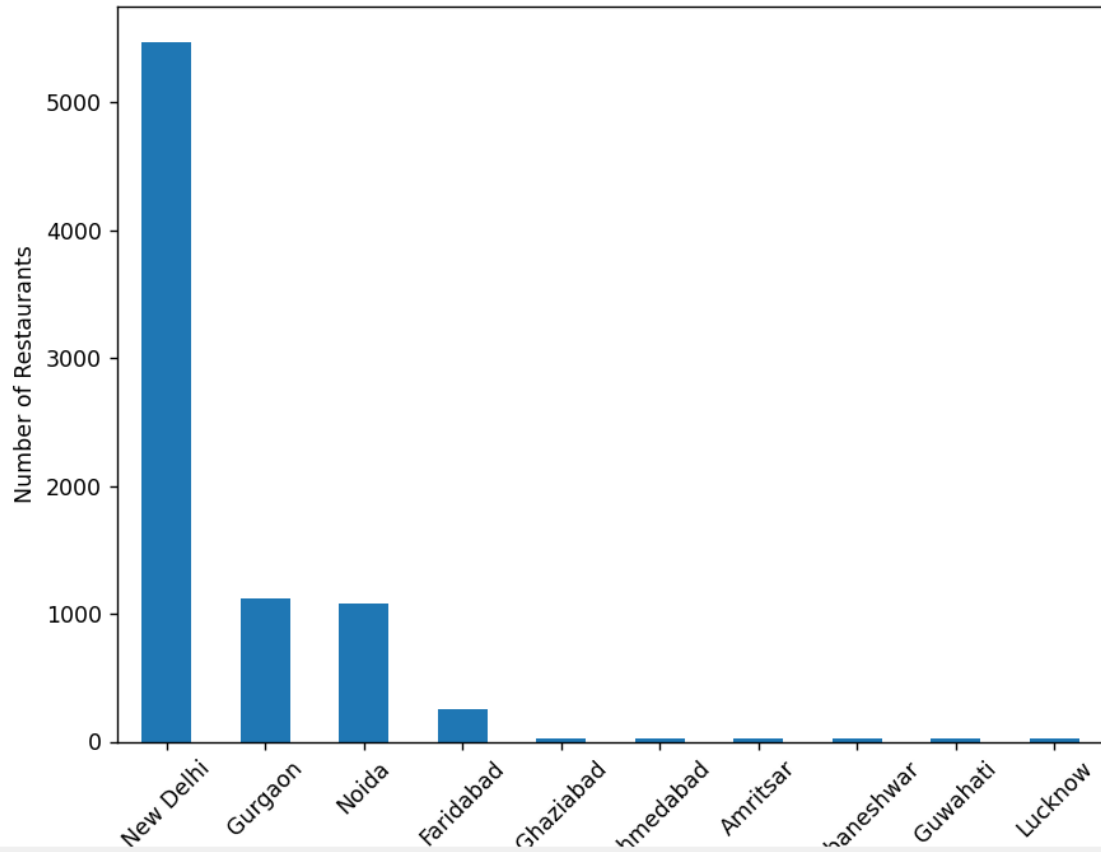
--- Top 10 Cities with Most Restaurants ---
City
New Delhi      5473
Gurgaon        1118
Noida          1080
Faridabad      251
Ghaziabad      25
Ahmedabad      21
Amritsar       21
Bhubaneswar    21
Guwahati       21
Lucknow        21
Name: count, dtype: int64

--- Longitude vs Aggregate Rating ---

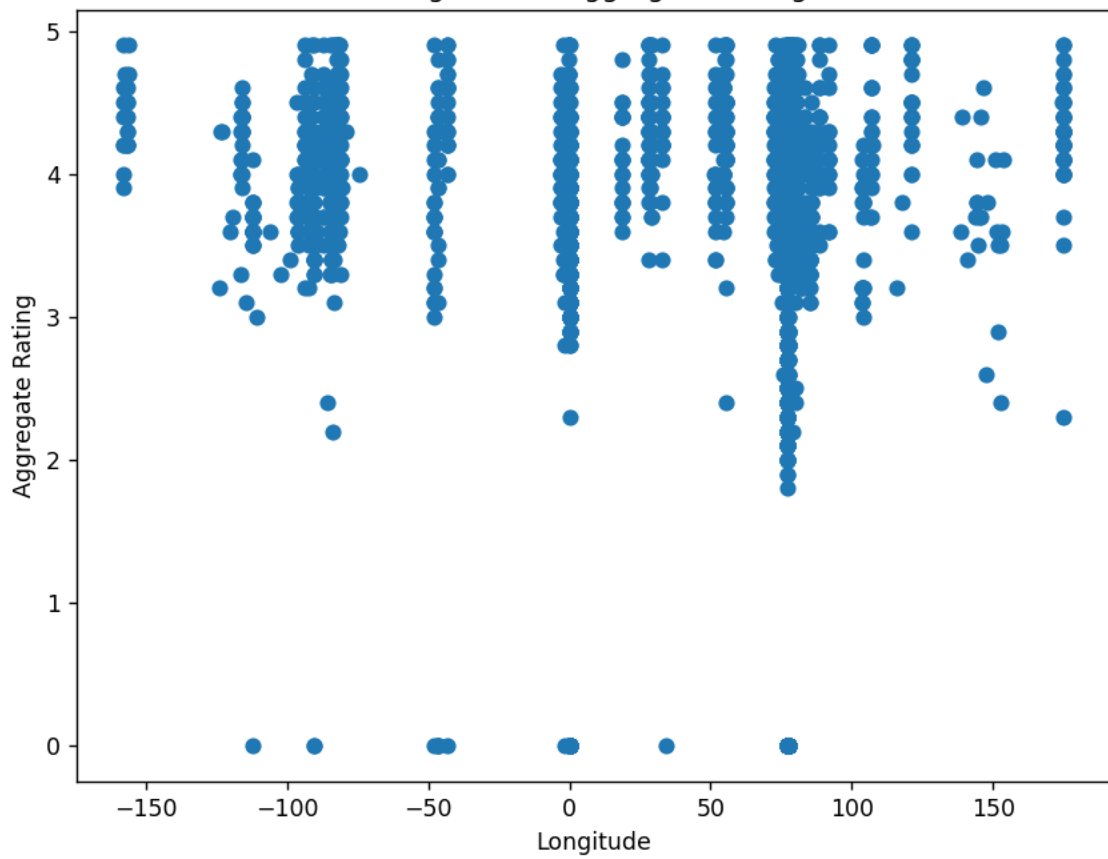
--- Latitude vs Aggregate Rating ---
```

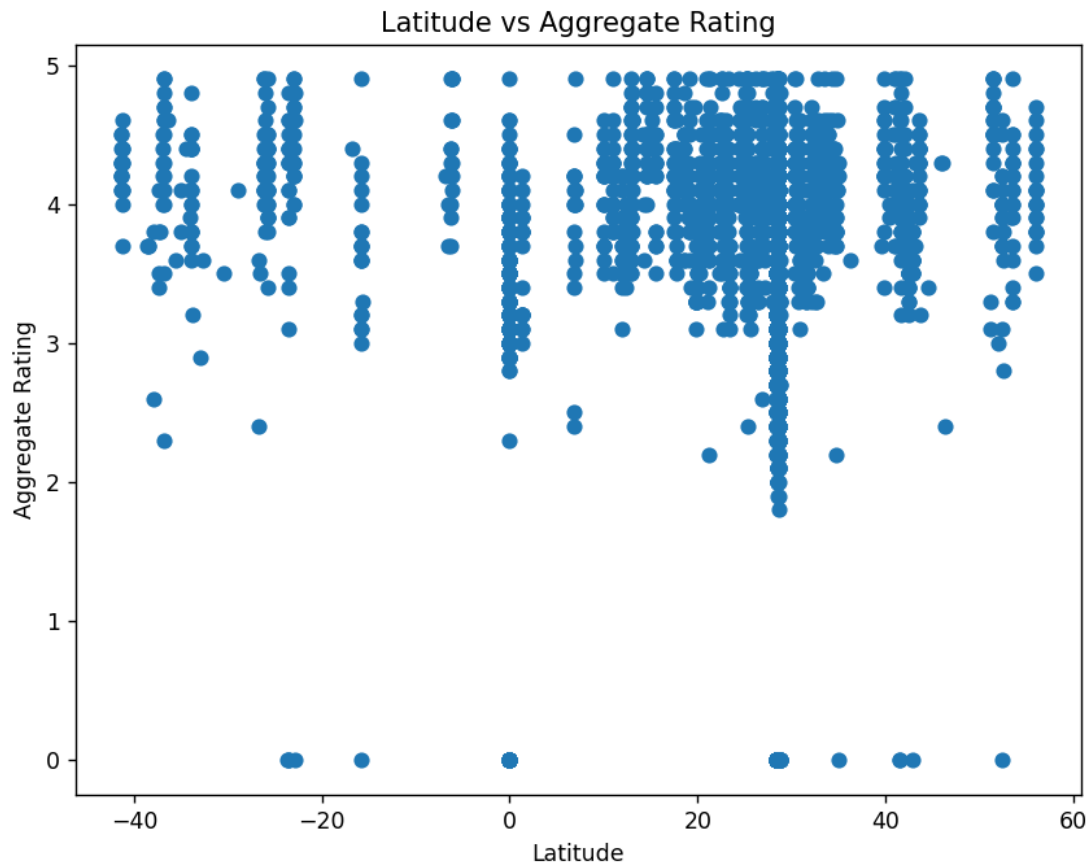


Top 10 Cities with Most Restaurants



Longitude vs Aggregate Rating





Visualization of Restaurant Locations :-

A scatter plot was created using latitude and longitude to visualize the geographical distribution of restaurants.

Observation:

- The restaurants are not evenly distributed across the map.
- A large number of restaurants are clustered in specific regions .
- The majority of restaurants appear concentrated in urban areas, especially around the Delhi NCR region.

Distribution Across Cities :-

The number of restaurants across different cities was analysed.

Key Findings:

- New Delhi has the highest number of restaurants
- Followed by Gurgaon and Noida
- Other cities have significantly fewer restaurants

👉 This shows that the dataset is heavily concentrated in a few major cities, particularly in the Delhi NCR region.

Relationship Between Location and Rating :-

Scatter plots were used to analyse the relationship between:

- Longitude and Aggregate Rating
- Latitude and Aggregate Rating

Observation:

- There is no strong or clear correlation between location and restaurant ratings
- Ratings are spread across all locations
- High and low ratings are found in similar geographic areas

Conclusion :-

The geospatial analysis reveals that restaurants are highly concentrated in specific urban regions, particularly in the Delhi NCR area. While location plays a role in restaurant distribution, it does not significantly influence the ratings of restaurants.

Overall, the analysis shows that restaurant ratings are independent of geographic location, and other factors such as service quality, cuisine, and pricing may have a greater impact on ratings.